



POSITIONING NAVIGATION AND TIMING INTERNSHIP



IIT Tirupati
Navavishkār
I-Hub Foundation



IIT Tirupati Navavishkar I-Hub Foundation &

Siddhartha Academy of Higher Education (Deemed to be University), India

“HANUMA”(HAZARD ANALYSIS NAVIGATION UNIT FOR MINE ASSISTANCE)

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Mentor

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The Critical Need for Automation in Hazardous Environments



Human Risk: Traditional manual inspection in underground mines exposes workers to fatal risks.



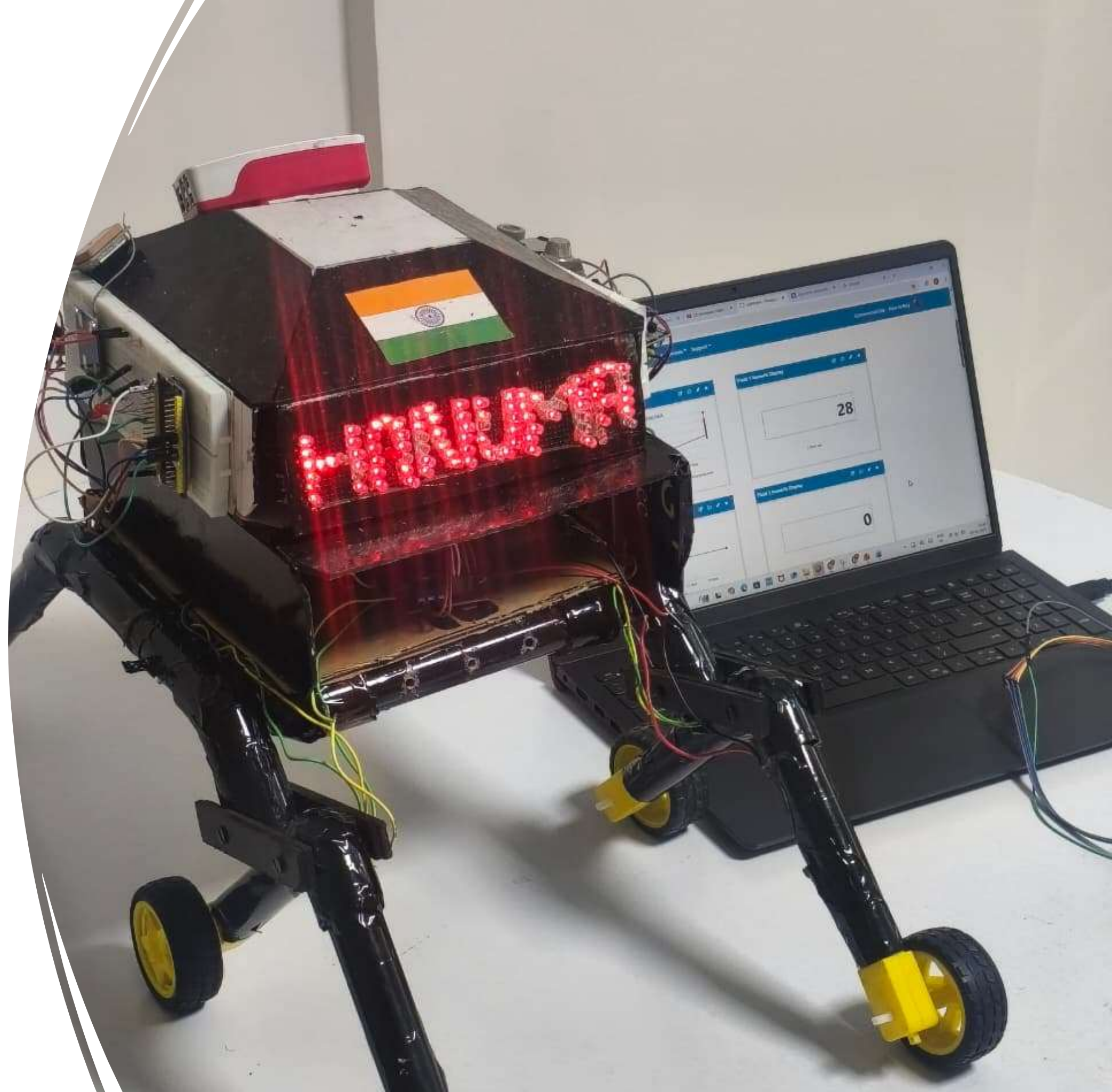
Environmental Dangers: Threats include toxic gas leaks (Nitrogen DiOxide, Carbon Dioxide), structural instability, and fire hazards from temperature spikes.



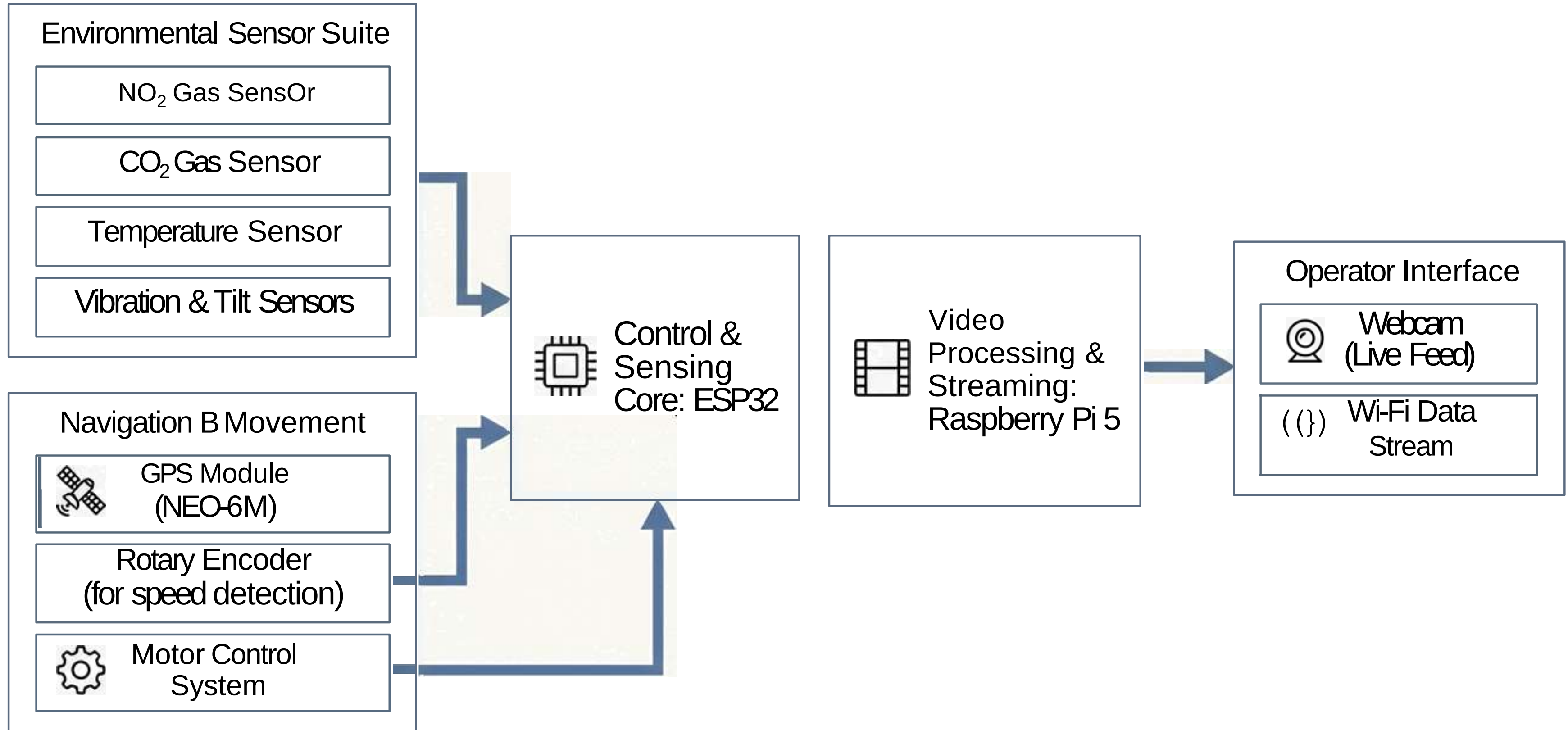
Data Gap: Safety teams lack real-time visual and sensory feedback from deep within tunnels before human entry.

Our Solution: A Robotic First Responder

- An autonomous rover designed to enter hazardous zones, stream real-time environmental data, and provide a live visual feed to a safe control center, ensuring areas are clear before human entry.

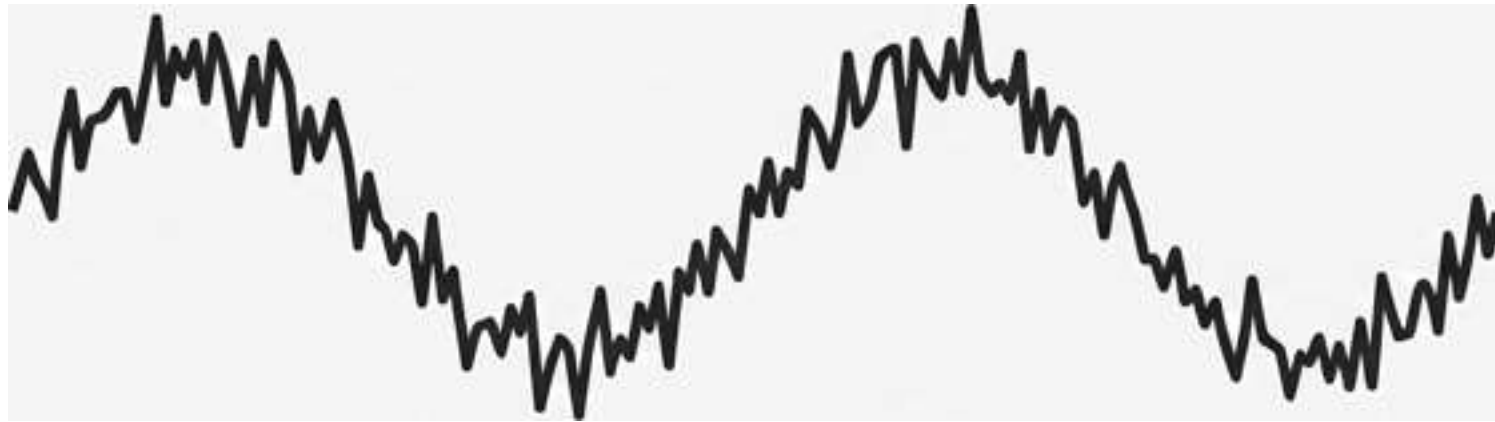


System Architecture



The Build, Part 1: From Raw Signals to Reliable Senses

Challenges

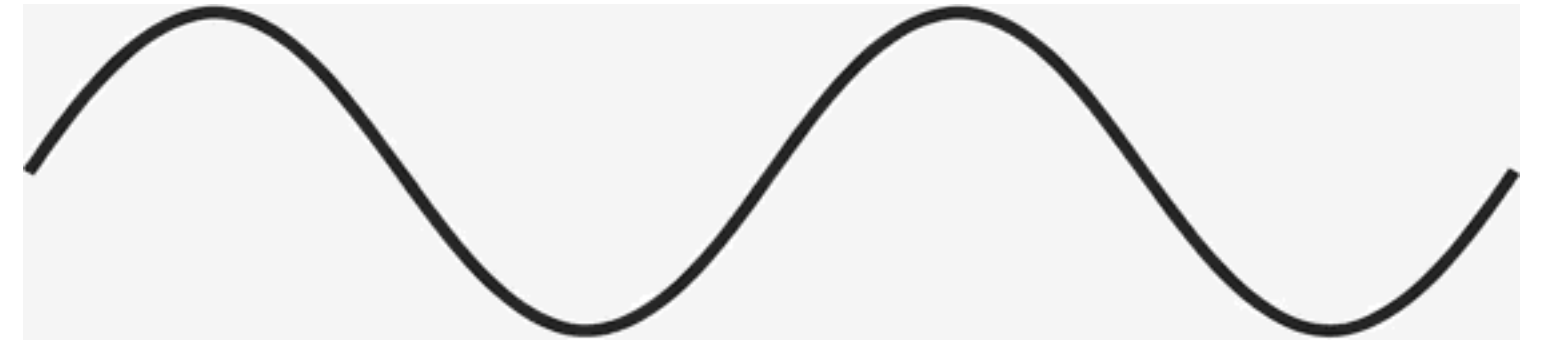


Gas Sensor Noise: Initial readings from NO₂ and CO₂ sensors were unstable and fluctuating.



Rotary Encoder Bounce: Mechanical bouncing generated multiple false pulses, corrupting speed detection.

Solutions



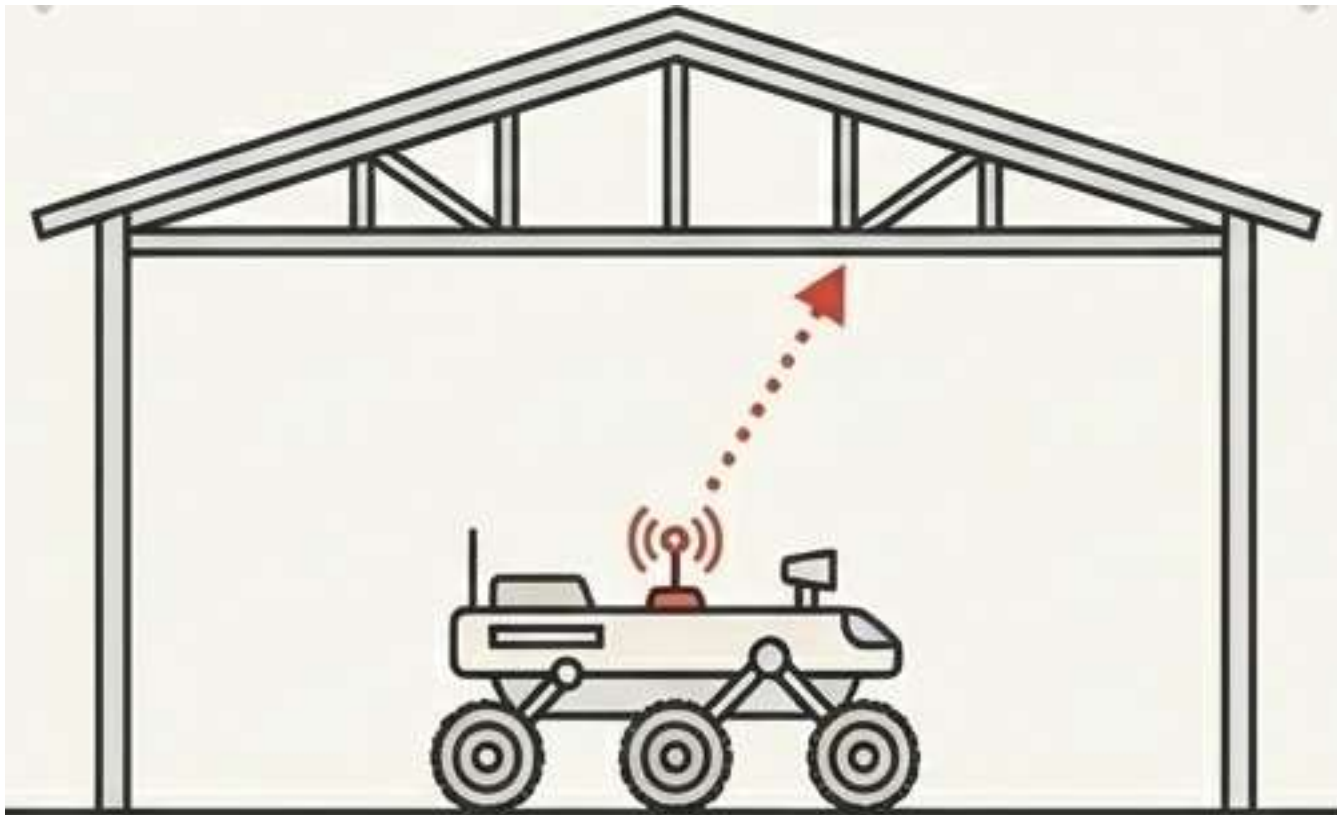
Data Filtering Implemented analog filtering and code-based smoothing algorithms to stabilize sensor output.



Signal Debouncing: Developed and optimized an Interrupt Service Routine (ISR) with debouncing logic to ensure accurate pulse detection from the encoder.

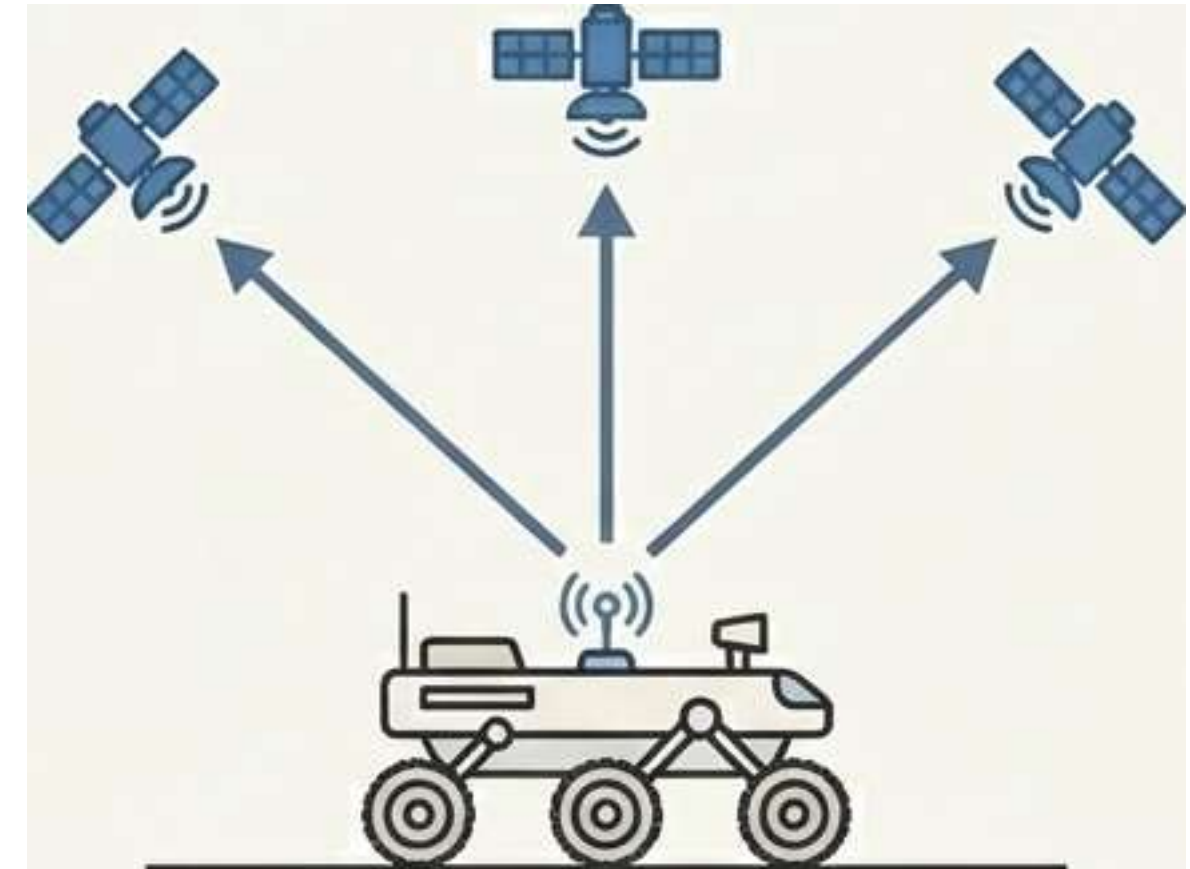
The Build, Part 2: Achieving Positional Awareness

Challenge



Satellite Acquisition: The NEO-6M GPS module experienced significant delays in achieving a satellite lock, hindering real-time location tracking.

Solution



Protocol & Placement: Diagnosed the issue as a signal problem. Improved satellite acquisition speed by strategically placing the antenna and establishing an outdoor testing protocol with a clear sky view. This also validated the use of the TinyGPS+ + library for parsing NMEA data via UART.

The Build, Part **3**: Integrating a Unified System

Challenges

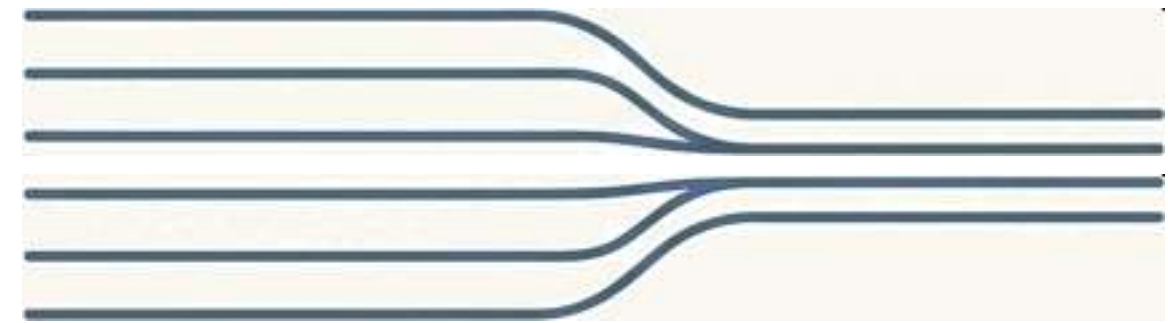


Processing Latency: Running multiple processes simultaneously (sensing, movement, communication) caused slight but noticeable delays in rover response.

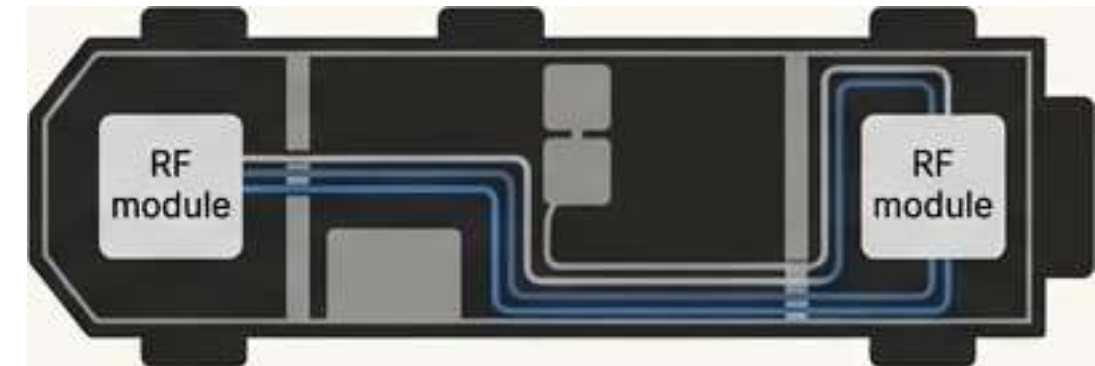


Signal Interference: The GPS and Wi-Fi modules interfered with each other, causing instability in both location data and the data stream.

Solutions



Code Optimization: Refactored the code to remove blocking delays and improve process handling, increasing responsiveness.



Hardware Isolation: Physically separated the wiring paths and optimized the placement of the RF modules on the chassis to reduce signal interference.

The Build, Part 4: A Foundation for Reliability

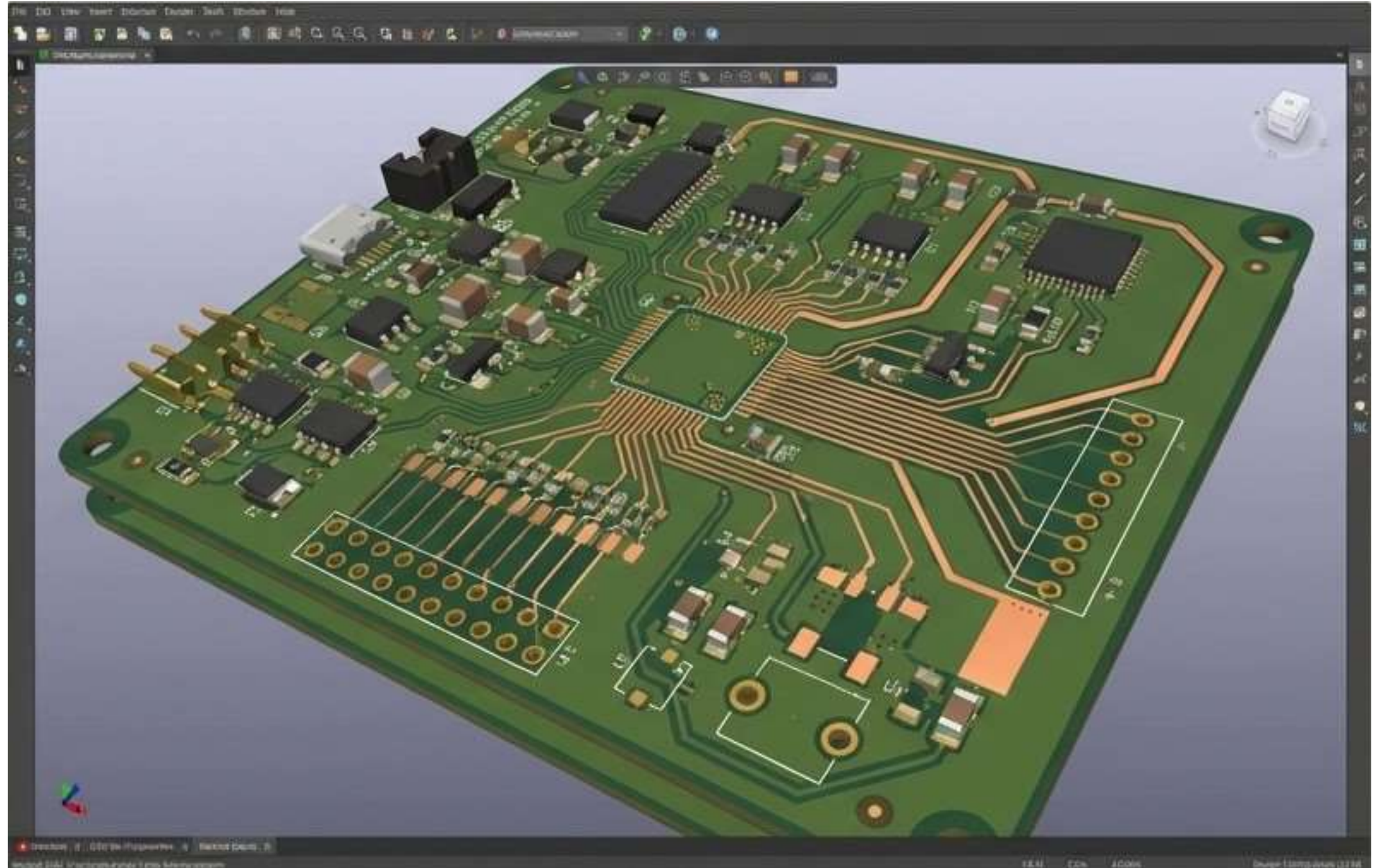
Objective: To create a robust, reliable motor-control subsystem, we designed a custom Printed Circuit Board (PCB).

Benefits:

Ensures stable performance under vibration.

- Reduces complex point-to-point wiring.
- Improves overall system reliability.

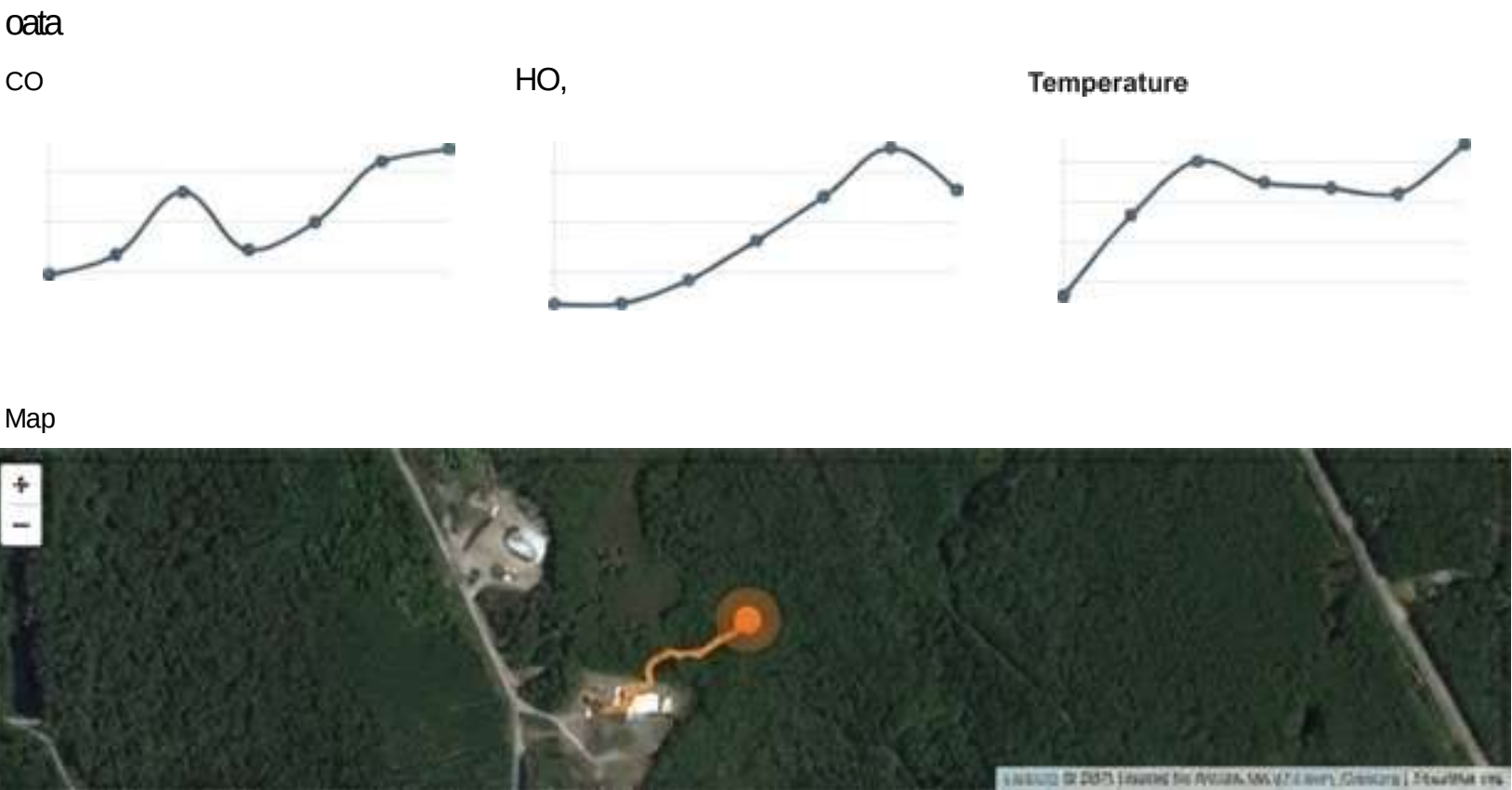
Status: Component placement is complete, routing is in progress.



The Result: Real-Time Situational Awareness

= Mission Control

Tn Live Video



Live HD Video Streaming



Continuous Environmental Monitoring

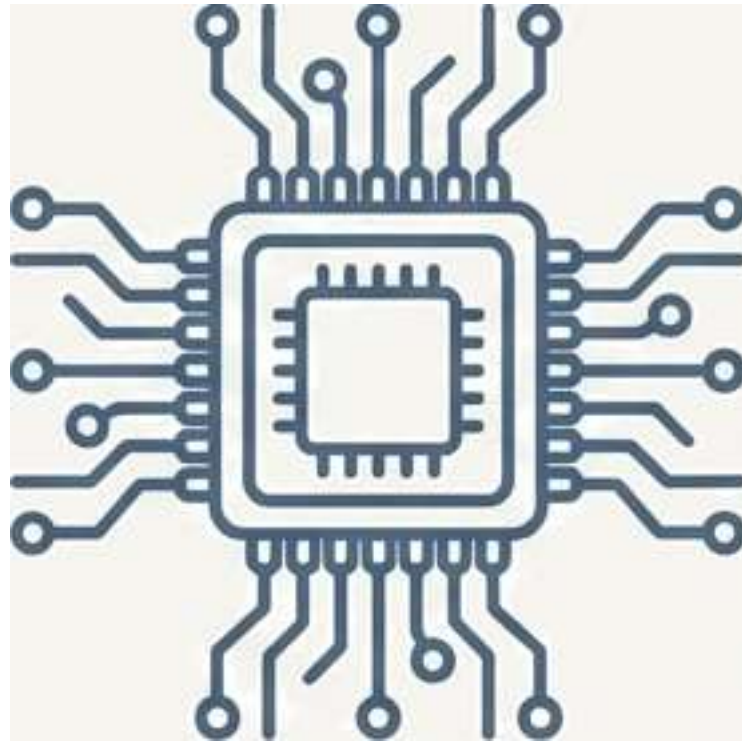


Precise GPS Location Tracking



Remote Rover Control

Key Learnings and Competencies



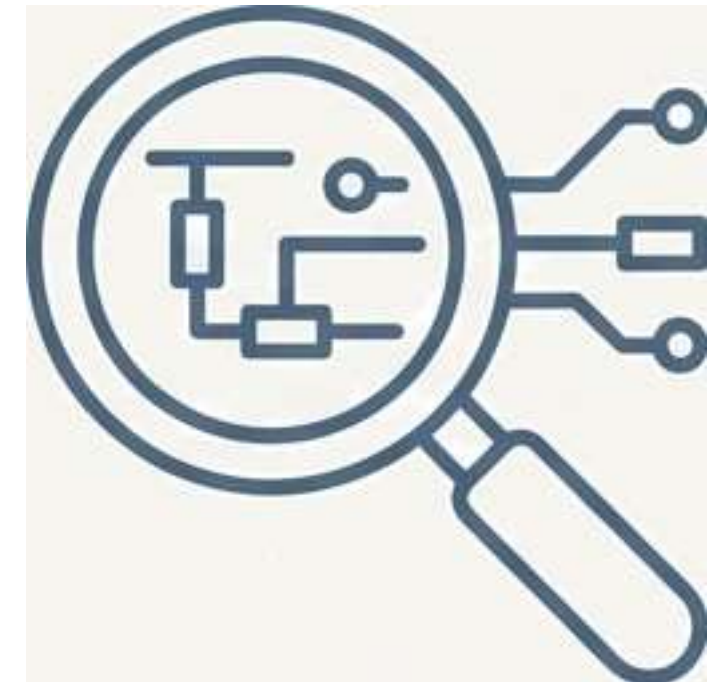
Hardware Integration

- Mastered sensor calibration, power management, and UART serial communication between modules.



Embedded Programming

- Gained proficiency in handling real-time inputs, interrupts, and multi-process optimization on the ESP32.



Systematic Troubleshooting

- Developed a rigorous methodology for diagnosing and solving complex hardware/software conflicts, from signal noise to RF interference.

Project Status and Future Trajectory

Immediate Next Steps

- Final Validation: Perform final, comprehensive testing of the fully integrated rover to verify reliable, simultaneous operation of all modules.
- Documentation: Complete and submit all project documentation, including schematics, code repositories, and final reports.

Future Vision

- Implement basic autonomy with on-board obstacle avoidance.
- Design a more robust, environmentally-sealed chassis for real-world deployment.

Acknowledgements

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Supporting Institution

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